

TBILISI INTL, Georgia

WMO: 375450

Lat: 41.667N

Lon: 44.950E

Elev: 495

StdP: 95.52

Time Zone: 4.00 (E04)

Period: 05-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-4.6	-12.1	1.4	-4.6	-9.8	1.7	-2.1	18.8	1.8	17.2	2.6	1.8	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	35.1	21.9	33.9	21.5	32.3	21.5	24.9	31.9	23.7	30.6	22.6	29.3	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.5	29.6	21.5	17.2	28.2	20.4	16.0	27.0	78.8	31.7	73.9	30.7	69.5	29.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.3	12.5	-8.4	36.9	3.0	1.7	-10.5	38.2	-12.2	39.2	-13.9	40.1	-16.0	41.3		
(5)			-9.5	26.4	2.9	1.0	-11.6	27.1	-13.3	27.7	-15.0	28.2	-17.1	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.4	3.6	8.3	13.0	18.4	23.1	25.7	25.9	21.2	15.6	8.9	4.2
	DBStd	8.94	3.41	3.94	3.52	3.95	3.14	2.42	2.56	2.92	3.00	2.83	3.49	3.22
	HDD10.0	746	234	179	76	14	0	0	0	0	0	1	59	182
	HDD18.3	2240	493	412	312	165	36	1	1	0	8	92	282	438
	CDD10.0	2302	0	1	22	104	262	394	487	494	334	174	27	2
	CDD18.3	753	0	0	0	4	40	144	229	236	93	7	0	0
	CDH23.3	6807	0	0	2	46	276	1152	2232	2458	610	32	0	0
	CDH26.7	2889	0	0	0	7	53	428	1022	1198	180	2	0	0
Wind	WSAvg	4.6	4.1	4.6	5.9	5.3	4.8	5.0	5.1	4.7	4.3	3.7	3.7	3.6
Precipitation	PrecAvg	479	19	24	30	52	75	72	43	45	34	37	31	20
	PrecMax	813	52	66	78	135	180	274	156	130	151	100	105	74
	PrecMin	254	0	0	1	12	5	21	1	2	0	0	2	0
	PrecStd	112	15	17	19	32	41	47	37	32	31	25	22	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.2	21.8	27.1	30.8	33.9	36.8	37.2	33.8	26.1	20.2	15.7
		MCWB	7.2	9.3	11.8	16.9	20.4	21.6	22.1	22.7	20.7	17.4	12.7	9.6
	2%	DB	10.8	13.6	18.9	24.5	28.0	32.2	34.9	35.8	30.9	24.0	18.1	12.8
		MCWB	6.4	7.8	11.2	15.7	19.0	21.2	21.8	21.9	20.3	17.0	12.3	8.2
	5%	DB	8.9	11.3	16.9	22.3	26.2	31.0	33.4	34.3	28.5	22.2	15.8	10.9
		MCWB	5.1	6.7	10.3	14.8	18.0	20.9	21.7	21.6	19.3	15.9	11.1	7.1
	10%	DB	7.1	9.2	14.3	20.0	24.6	29.3	32.0	33.0	26.8	20.8	14.0	9.2
		MCWB	4.3	5.9	9.0	13.7	17.5	20.6	21.9	21.6	18.9	15.1	10.3	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	10.0	13.7	18.2	21.4	23.6	25.7	26.9	23.8	19.1	14.2	10.4
		MCDB	11.5	15.1	20.4	24.9	28.7	31.3	33.4	33.8	30.5	23.8	18.7	14.1
	2%	WB	6.9	8.4	11.8	16.7	19.8	22.5	24.4	25.5	21.7	17.6	13.0	9.0
		MCDB	9.5	12.7	17.3	22.9	26.6	30.1	32.1	32.8	27.8	22.1	16.0	12.0
	5%	WB	5.6	7.2	10.5	15.4	19.0	21.8	23.5	24.2	20.6	16.7	11.9	7.8
		MCDB	8.2	10.8	15.6	20.7	25.3	29.2	30.8	30.6	26.3	20.8	14.3	10.4
	10%	WB	4.5	6.1	9.5	14.2	18.0	21.0	22.6	22.9	19.6	15.8	10.9	6.4
		MCDB	6.7	8.8	13.7	18.9	23.4	27.8	29.3	29.1	25.0	19.4	13.5	8.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.9	9.4	9.6	10.0	10.8	10.9	11.5	9.9	8.8	8.2	7.9
		MCDBR	10.2	11.6	13.8	13.4	13.0	13.2	13.3	13.9	12.3	11.3	11.6	10.4
		MCWBR	7.3	7.5	8.2	7.5	6.9	6.2	6.6	6.5	6.5	6.2	6.6	6.8
	5% WB	MCDBR	9.1	10.8	11.9	12.4	11.7	12.2	12.0	12.5	10.8	10.2	9.6	9.2
		MCWBR	6.8	7.1	7.5	7.2	6.4	6.2	6.4	6.8	6.5	6.0	6.0	6.2
Clear-Sky Solar Irradiance	taub	0.332	0.369	0.415	0.480	0.449	0.439	0.481	0.475	0.424	0.401	0.353	0.330	
	taud	2.327	2.211	2.118	1.977	2.139	2.226	2.119	2.129	2.254	2.254	2.348	2.347	
	Ebn at Noon	801	820	822	793	832	839	798	790	806	775	770	769	
	Edh at Noon	87	114	140	174	151	139	153	148	122	108	85	79	
All-Sky Solar Radiation	RadAvg	1.86	2.72	3.48	4.37	5.31	6.21	6.06	5.55	4.32	2.87	2.02	1.61	
	RadStd	0.22	0.38	0.38	0.59	0.38	0.43	0.50	0.57	0.36	0.32	0.23	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page